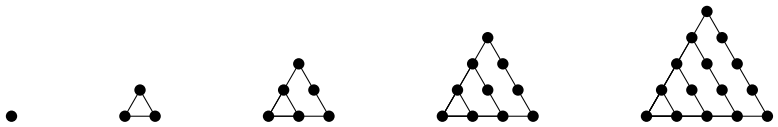
All this fires my soul, and provided I am not disturbed, my subject enlarges itself, becomes methodised and defined, and the whole, though it be long, stands almost complete and finished in my mind, so that I can survey it, like a fine picture or a beautiful statue, at a glance.
—Wolfgang Amadeus Mozart

PROMYS Number Theory **Open Door Problem Set** *Boston University, August 8, 2025*

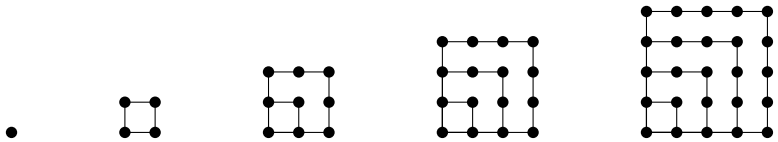
During the past six weeks we have investigated a number of fundamental mathematical themes. We hope you have found your experience stimulating and that you will choose to continue your studies. The problems below are all quite challenging. In fact, one of them – the last question in P8 – is an unsolved problem of modern mathematics. You will probably find that the books on our reading list will help you not only with your work on these problems but also in your future mathematical studies. Please let us hear from you, perhaps in December, about your ideas on the problems or any other matters which you would like to share with us. We hope to see many of you again next summer. Best wishes for the coming year. – Glenn Stevens

P1: The triangular numbers are the numbers $1, 3, 6, 10, 15, \ldots$; the square numbers are $1, 4, 9, 16, 25, \ldots$; the pentagonal numbers are $1, 5, 12, 22, 35, \ldots$. The geometrical language is justified by the following diagrams:

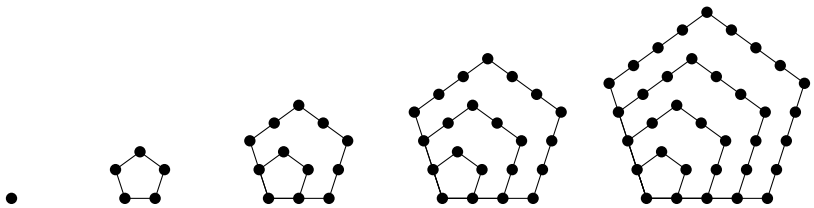
Triangular numbers:



Square numbers:



Pentagonal numbers:



What are the first five hexagonal numbers? What are the first five septagonal numbers? What are the first five r-gonal numbers? Give a formula for the nth triangular number. Give a formula for the nth square number. Give a formula for the nth pentagonal number. In general give a formula for the nth r-gonal number.

P2: Every positive integer is either a triangular number or can be expressed as a sum of either 2 or 3 triangular numbers. Every positive integer can be expressed as a sum of 4 or fewer square numbers. Also every positive integer can be expressed as a sum of 5 or fewer pentagonal numbers. In general, every positive integer can be expressed as a sum of r or fewer r-gonal numbers.

P3: There are only finitely many integral solutions $m, n \in \mathbf{Z}$ to the equation

$$2^n - 3^m = 1.$$

More generally, if p and q are primes then

$$p^n - q^m = 1$$

has at most finitely many solutions $m, n \in \mathbf{Z}$. Can you generalize these assertions?

P4: Given a prime p, then there exist irreducible polynomials in $\mathbf{Z}_p[x]$ of every degree. For a given positive integer n, let

$$P_n(x) = \prod_f f(x)$$

where the product is over all irreducible monic polynomials of degree n. What is $P_1(x)$? For general n, we have the following polynomial identity in $\mathbf{Z}_p[x]$:

$$\prod_{\substack{d|n \\ d>0}} P_d(x) = x^{p^n} - x.$$

P5: Consider the equation $y^2 = x^3 - x$. For each prime $3 < p < 20$ calculate the number, N_p, of solutions of this equation in $\mathbf{Z}_p$. For example, $N_5 = 7$ since there are seven solutions of $y^2 = x^3 - x$ in $\mathbf{Z}_5$, namely $(x, y) = (0, 0)$, $(1, 0)$, $(2, 1)$, $(2, 4)$, $(3, 2)$, $(3, 4)$, and $(4, 0)$. In general, we have the following formula for N_p.

$$N_p = \begin{cases} p & \text{if } p \equiv 3 \pmod 4 \\ \\ p - 2a & \text{if } p \equiv 1 \pmod 4 \text{ and } p = a^2 + b^2 \text{ where } \pi = a + bi \text{ is a prime} \equiv 1 \text{ modulo } 2 + 2i \text{ in } \mathbf{Z}[i]. \end{cases}$$

Can you fomulate and prove a conjecture concerning the number of solutions in $\mathbf{Z}_p$ to the equation $y^2 = x^3 - Dx$ for arbitrary non-zero integers D?

P6: The number of elements in a finite field is a prime power, p^n. For every prime p and every positive integer n there is a field with p^n elements. If two finite fields have the same number of elements, then they are *isomorphic*. Every finite field has a primitive root.

P7: Find all the integral solutions $x, y \in \mathbf{Z}$ of the following Diophantine equations.

(a) $y^2 = x^3 - 2$;

(b) $y^2 = x^3 + 7$;

(c) $y^2 = x^3 + 1$.

In each case, also find all the rational solutions $x, y \in \mathbf{Q}$.

P8: Given a square-free integer d, let $R_d = \{\, a + b\omega_d \mid a, b \in \mathbf{Z} \,\}$ where

$$\omega_d = \begin{cases} \dfrac{1 + \sqrt{d}}{2} & \text{if } d \equiv 1 \pmod 4; \\ \\ \sqrt{d} & \text{otherwise.} \end{cases}$$

Then R_d is closed under multiplication and addition and is a ring with these two operations. There are only finitely many values of d for which R_d has a division algorithm. What are they? The ring R_{-19} does not have a division algorithm. Nevertheless, it is true that the linear Diophantine equation $aX + bY = g$ is solvable whenever g is a GCD of a, b in R_{-19}. Can you prove this? As we have seen many times this summer, the fundamental theorem of arithmetic is a consequence of the solvability of this equation. Hence, the fundamental theorem holds in R_{-19} even though this ring has no division algorithm. Conclude that for a prime $p > 2$ in $\mathbf{Z}$, the Diophantine equation $x^2 - xy + 5y^2 = p$ has an integral solution (x, y) if and only if -19 is a quadratic residue modulo p. Can you generalize these ideas? Do you think that the fundamental theorem holds in R_d for all square free d? Does the fundamental theorem hold in R_d for infinitely many d? (See Hardy and Wright chapters XIV and XV for a discussion of some of these questions.)